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An empirical evidence on the impact of social customer relationship management on the small and medium enterprises performance

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ABSTRACT

Social media has swiftly established itself as a primary source of products' information for customers. Nowadays, Small and Medium-size Enterprises (SMEs) can use social media to develop Customer Relationship Management platforms (social CRM). Firms, particularly SMEs in developing countries need to understand the factors affecting their performance by implementing social CRM. However, there is a dearth of awareness on the impact of social CRM on the performance of SMEs. This study proposes an integrated model that aims to investigate the effects of social CRM on SMEs' performance. The model is constructed by incorporating three dominant theoretical frameworks: the Fit-Viability Model (FVM), Network Externalities, and the Resource-Based View (RBV). A cross-sectional survey was used to gather data from 149 SMEs managerial staff. Findings revealed that almost 50% of the variability in the performance of SMEs is explained by the fitness and viability of social CRM. In addition, network externalities of social media significantly impact the social CRM fitness in the context of SMEs with path coefficient 0.617. Furthermore, the internal financial resources factor makes sCRM viable for SMEs as the results show significant relationship between the internal financial resources and the sCRM viability with 0.536 path coefficient. Manager innovativeness, IT knowledge, top management support, and government assistance, on the other hand, do not contribute significantly to the viability of social CRM for SMEs. The model aids SMEs in making well-informed decisions regarding the adoption of social CRM by evaluating both the suitability of social media for CRM tasks and the enterprise preparedness to implement social CRM, leading to enhanced performance.

1. Introduction

Social media is becoming a fundamental aspect of daily life, serving as a means of communication and information exchange. It has introduced new ways for businesses to communicate with customers and connect with people using many-to-many communication techniques rather than one-to-many communication techniques (AlQershi, Mokhtar & Abas, 2020; Shazad et al., 2020). Small and medium-sized enterprises (SMEs) may lack resources in terms of finance and expertise and may take advantage of social media since it is free and straightforward to use. According to Chatterjee and Kar (Chatterjee & Kar, 2020), adopting social media is a vital strategy for SMEs, particularly in developing countries. The literature asserted that adopting social media by SMEs has several benefits: including broader reach, improving knowledge sharing, access to global markets, and prompting communication with customers (Abed, 2020; Geurin & Burch, 2017; Jung & Jeong, 2020; Lamrhari, El Ghazi, Oubrich & El Faker, 2022; Parveen, Jaafar & Ainin, 2016; Tajudeen, Jaafar & Ainin, 2018). Social media enables enterprises and customers to communicate effectively, allowing firms to understand and react proactively and adequately to customers' needs (Parveen et al., 2016). As a result, SMEs can preserve competence and boost their performance by strengthening customer interactions (Wardati & Mahendrawathi, 2019). Indeed, social media offers a multi-directional communication mode that empowers firms to engage in interactions with both their existing customers and potential customers within their networks. This two-way communication allows companies to not only

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disseminate information to their target audience but also receive feedback, comments, and inquiries from customers and prospects. By leveraging social media platforms, businesses can establish direct and real-time connections with their audience, fostering meaningful relationships and building brand loyalty. Moreover, the ability to engage with customers and their networks allows firms to tap into new customer segments and extend their reach to a wider audience, potentially converting individuals from these networks into loyal customers (A.O. Banwo, Du & Onokala, 2017; Lehmkuhl, 2014; Shin, Oh2 & Jeong, 2020). Using social media to manage and maintain the firm's customer relationship is referred to as social CRM (or sCRM) (Choudhury & Harrigan, 2014).

In addition, the SME sector has significantly been affected by the COVID-19 pandemic. The ever-increasing outbreak of the corona epidemic has caused a significant halt in the business process. Accordingly, sales declined, eventually forcing many SMEs to close their businesses (Effendi, Sugandini & Istanto, 2020). On the other hand, online shopping is growing in popularity as it has become the primary choice for many individuals. Thus, small businesses are forced to utilize the Internet as a platform during times of crisis. When it particularly comes to engaging with customers, social media has emerged as an effective and efficient option (Effendi et al., 2020). However, to effectively use social media, SMEs should identify the potentials, drivers, and determinants of adopting these tools to enhance performance. Thaha, et al. (Thaha, Maulina, Muftiadi & Alexandri, 2021) conducted a bibliometric analysis of the studies published from 2009 to 2020 on social media in SMEs and concluded that three main topics were investigated: the adoption of social media in SMEs, the role of social media in SME innovation, and entrepreneurship, and the function of social networks in SMEs marketing. While many studies have explored social media acceptance and its use from the business-consumer perspective, there is a gap in the existing literature regarding how social CRM impacts SMEs' performance (Acker, Gröne, Akkad, Pötscher & Yazbek, 2011; Choudhury & Harrigan, 2014; Harrigan & Miles, 2014; Thaha et al., 2021). Thus, this research addresses the following questions:

1) What factors influence the performance of SMEs through the adoption of social CRM?
2) How can an integrated model be developed to leverage the relationships between the influencing factors?

The current literature may lack comprehensive coverage of the crucial factors that SMEs must consider while allocating resources for the implementation of social CRM (A.O. Banwo et al., 2017). It is vital for SMEs to be cognizant of the elements that influence their performance through sCRM (Çakıcı, 2022; Marolt, Pucihar & Zimmermann, 2015). Moreover, studies identifying social CRM performance results encourage more empirical studies on social CRM performance outcomes (Aldoseri, Al Mubarak & El Hajjar, 2019; Diffley, McCole & Carvajal-Trujillo, 2018; Effendi et al., 2020; Hassan, Mohamed Haniba & Ahmad, 2019; Küpper, 2016; Marolt et al., 2015). Therefore, this study aims to enhance the understanding of social CRM's impact on SMEs' performance in developing countries by proposing an integrated model that identifies the contributing factors.

This study contributes to the existing literature on information systems by extending the Fit-Viability Model (FVM), an established IS theory that models the impact of task-technology fit and the viability of technology initiatives on business performance. The extension of this model incorporates the network externalities theory to analyze the technology under investigation, particularly social media. Additionally, it integrates the Resource-Based View (RBV) theory to address the unique context of small and medium-sized enterprises (SMEs). By incorporating the network externalities theory, the study enhances our understanding of how social media technologies create value through the interconnectedness and interactions with customer. Furthermore, by incorporating the RBV theory, the study acknowledges the importance of considering internal resources and capabilities within SMEs to effectively adopt and utilize social CRM (Customer Relationship Management).

The study emphasizes the practical implications of social CRM, particularly within the context of SMEs. It offers valuable guidance for SMEs seeking to harness the power of social media to manage and enhance customer relationships effectively. By leveraging social CRM strategies, SMEs can gain a competitive advantage and achieve improved overall performance in today's digital landscape.

The subsequent sections of this paper are structured as follows: Next section presents the research theoretical background, Section III proposes the research model constructs and hypotheses, Section IV outlines the research methodology employed to answer the research questions, Section V presents the analysis and the empirical results, subsequently, Section VII discusses the findings and finally, Section VIII provides a conclusion of the research.

2. Theoretical background

The assessment of the impact of IT innovation on organizational performance is a challenging endeavor, necessitating the integration of multiple theoretical paradigms (Melville, Kraemer & Gurbaxani, 2004). Researchers have employed various theories to better comprehend the factors linked to the utilization of ICT in organizations (Kling, 1980; Markus & Robey, 1988). Given the focus of this research on SMEs and their adoption of sCRM, it is crucial to consider theories or models that specifically address performance measurement within the context of SMEs and social media. One such model is the Fit Viability Model (FVM), which extends the Task-Technology Fit (TTF) approach. FVM identifies the factors that predict technology usage and its subsequent impact on organizational performance (Baas & Rekom, 2010; Tjan, 2001). By incorporating theories that align with the unique characteristics of SMEs and the dynamics of social media, this study aims to gain a comprehensive understanding of the implications of sCRM adoption on SME performance.

Deploying innovative technology in an organization can entail a certain level of risk. Hence, possessing a model capable of foreseeing the successful implementation of such technology within the organization's distinct context can offer significant benefits. This forecast is based on evaluating both the organization's preparedness and the specific features of the technology being adopted (Mohammed, Ibrahim, Nilashi & Alzurqa, 2017). FVM can be a valuable tool in determining whether adopting sCRM can indeed enhance the performance of SMEs. It takes into account two critical aspects: technology features (fitness) and organizational readiness (viability) (T.P. Liang, Huang, Yeh & Lin, 2007). In the context of this study, "Fit" assesses how well sCRM aligns with the specific business tasks of SMEs. On the other hand, "Viability" pertains to the value sCRM can bring to a company's performance and how prepared the organization is to implement it. By evaluating these factors, FVM aids in gaging the potential success of sCRM in SMEs and its positive impact on their performance.

Furthermore, the Resource-Based View (RBV) theory and network externalities have garnered considerable recognition and validation in the social CRM literature (Ahani, Rahim & Nilashi, 2017; A.O. Banwo et al., 2017; Lehmkuhl, 2014; Trainor, Andzulis, Rapp & Agnihotri, 2014; Wittkuhn, Lehmkuhl, Küpper & Jung, 2015). As per network externalities theory, the value of a technology increases with the size of its user network. This holds true for sCRM, given that social media platforms facilitate the participation of a large user base (A.O. Banwo et al., 2017; V. Gu, Davis, Cao & Vogt, 2017). Consequently, the social network externality can significantly impact the suitability of social CRM for SMEs' task needs, particularly in terms of fostering strong customer involvement.

Additionally, RBV theory highlights that firms can gain a competitive edge by having valuable, rare, and unique resources that cannot easily be replaced (Barney, 1991). Within the realm of SMEs, integrating

resources like human capital, technology, and business competencies can be utilized to cultivate CRM capabilities that elevate overall performance (Rapp, Trainor & Agnihotri, 2010; Trainor et al., 2014). Notably, several resources have been identified in the literature to influence firms' readiness to adopt sCRM, including Internal Financial Resources (Hasani, Bojei & Dehghantanha, 2017; Sophonthummapharn, 2009), Manager Innovativeness (Hasani et al., 2017; Parveen et al., 2016), IT/IS Knowledge of Employees (Ahani et al., 2017), Top Management Support (Ahani et al., 2017), and Governmental Support (Ahani et al., 2017; Hasani et al., 2017).

Therefore, the study proposes a relationship between the RBV and the "viability" component of the FVM model. By integrating FVM, network externalities, and RBV, an encompassing model is developed to explore the various factors influencing the adoption of social CRM and its potential impact on enhancing SME performance in developing countries. This integrated model allows for a comprehensive analysis of the complex dynamics involved when implementing social CRM in SMEs' organizational contexts.

3. Proposed model constructs and hypotheses

Based on the proposed model, Network Externalities as a characteristic of social media can influence the performance of sCRM to the requirements of SMEs, which can affect the firm's performance. In addition, factors including manager innovativeness, IT knowledge, internal financial resources, top management support, and government support can contribute to the viability of social CRM in SMEs, which can affect the firm's performance.

3.1. Network externalities

Network externalities is a characteristic that describes the increase of the value of technology when the number of users increases. The more time users spend on a given social site, the broader their social networks have, which produces greater value for the users (Hong, Cao & Wang, 2017; Katz & Shapiro, 1985). This is true for CRM since social media technologies enable the interaction of a large number of users (A.O. Banwo et al., 2017; V. Gu et al., 2017). Therefore, we argue that network externalities might impact the suitability of social CRM to the needs of SMEs in managing their customer interactions. As a result, this research hypothesizes:

H1. Network externalities positively influence the fitness of social CRM to SMEs' requirements.

3.2. Manager innovativeness

Innovativeness refers to humans' capacity to engage in innovation (Hult, Hurley & Knight, 2004). Innovation for small business managers implies learning and adopting innovation in their business (Alam & Dubey, 2014). Firms can significantly improve their results if they innovate their business processes through technological innovations (Leoncini, 2016). Managerial innovation has a positive impact on SMEs' use of various technologies to manage firms' relationships with customers (Potocan & Nedelko, 2013), including electronic CRM (eCRM) (Ghobakhloo & Tang, 2013), mobile CRM (mCRM) (Luo, Zhang & Duan, 2013), and even social networking (Hasani et al., 2017). Innovative Managers are more likely to devise many departure options, affecting the company's capacity to embrace durable technology like new CRM systems. Thus, managers' creativity is a company competency that influences an organization's preparedness to implement social CRM. It can be inferred that managers are likely to implement various strategies that could impact the company's capacity to adopt long-lasting technologies, such as new customer relationship management solutions.

Accordingly, managers' innovativeness can be considered a firm capability that affects the firm readiness to adopt social CRM. Thus, the following hypothesis is proposed:

H2. Managers' innovativeness positively impacts the viability of sCRM in the context of SMEs.

3.3. IT knowledge

Employees are assets since firms' success relies on them. Employees' IT expertise helps firms to maximize resources for adopting IT innovations. Accordingly, IT knowledge is a firm's capital to absorb changes and adapt to the environment (Nguyen, 2009). Firms with knowledgeable IT personnel may wish to investigate new technologies and may be willing to take risks to adopt the technology (Ramdani, Kawalek & Lorenzo, 2009). Many studies explored IT expertise as an influential factor contributing to the adaptation of technology in the context of SMEs (Alam & Noor, 2009; Hassan, 2017; Miraz & Habib, 2016). In particular, numerous studies examined workers' IT competence as a predictor of social media adoption by SMEs (Ahani et al., 2017; Effendi et al., 2020). Therefore, the following hypothesis might be proposed:

H3. IT knowledge positively influences the viability of sCRM in the context of SMEs.

3.4. Top management support

Ifinedo (Ifinedo, 2011) defines top management support as management engagement and encouragement to implement information systems. Top management influences members' acceptance of new technologies. A favorable atmosphere and resources for technology adoption need top management support (Premkumar & Roberts, 1999). Top management support is one of the most significant predictors of organizational adoption of information system innovation (Jeyaraj, Rottman & Lacity, 2006). Much additional research has linked top management support to an organization's chance of embracing social media (Ahmad, Bakar & Ahmad, 2019; Effendi et al., 2020; Sugandini, Effendi & Istanto, 2020). Thus, Top management support can be a predictor of the success of social media in managing customer relationships. Hence, this research proposes:

H4. Top management support positively impacts the viability of sCRM in the context of SMEs.

3.5. Internal financial resources

Adopting new technology demands financial investment for installation, training, and adaptation (Kim & Shin, 2015). Firms with more outstanding internal funds embrace new technology faster (Ghobakhloo & Hong, 2014). Thus, limited capital is a barrier that prevents SMEs from benefiting from new technologies (Tan, Chong, Lin & Eze, 2009). According to Kurnia, et al. (Kurnia, Choudrie, Mahbubur & Alzougool, 2015), financial resources are essential to adopt a technology. For SMEs, social networking platforms and sCRM solutions are affordable (Kirtiş & Karahan, 2011), which makes them viable for SMEs. Thus, this study postulates that:

H5. Financial resources positively influence the viability of sCRM in the context of SMEs.

3.6. Government support

Government support is provided through policies and programs that encourage and facilitate the shifting to new technologies (Gibbs & Kraemer, 2004; Hadi, Omar & Osman, 2021). Government support is crucial for SMEs to use the latest technology. Government IT strategies should assist firms in overcoming challenges and accepting innovative IT (Faqir, Sharari & Salameh, 2014). Aljowaidi (Aljowaidi, 2015) claimed that a lack of government support hinders the implementation

of e-commerce by SMEs. Furthermore, government support for social media can either prompt or deter firms' utilization of social networking technologies (Park, Choi, Kim & Rho, 2015). Previous research has shown government support's impact on businesses' social media adoption (Kim, Park & Rho, 2015). Government support for social media adoption has boosted sCRM among firms and added more value (Parveen, Jaafar & Ainin, 2015). Thus, government support can assist and promote sCRM adoption in SMEs. Therefore, this study proposes the following hypothesis:

H6. Government support contributes positively to the viability of sCRM for SMEs.

3.7. Fit

The fitness of a technology characteristics with an organization's core competencies and needs is called task technology fit or "fit" (T.P. Liang, Huang, Yeh & Lin, 2007). Lippert and Forman (Lippert & Forman, 2006) also defined fitness as the degree to which technology delivers features and meets task requirements. According to Liang et al. (T.P. Liang et al., 2007), fitness measures how well technology characteristics meet the demands of a job. In the present research context, fitness is defined as the amount to which social media is congruent with the necessity of waging the SME customer's interactions. Fit has previously been investigated in a variety of scenarios, including electronic government (Larosiliere & Carter, 2016), cloud computing (Tripathi & Nasina, 2017), and Blockchain (Liang, Kohli, Huang and Li, 2021). Generally, research in the literature demonstrates that when technology features match the context of work needed, the organizations' performance is enhanced. Therefore, the following hypothesis is proposed:

H7. The fitness of sCRM to the SEMs' requirements positively influences their performance.

3.8. Viability

The viability of a technology is measured by the potential additional value it provides, including human resources, infrastructure, and financial benefits of adopting the technology (T.P. Liang et al., 2007). Depending on the circumstances, several variables impact an organization's viability to embrace new technology. Liang and Wei (Liang & Wei, 2004) examined economic feasibility, senior management support, and IT infrastructure to determine viability. To make sCRM a viable option for SMEs, it may need more informed users and top-level assistance, as well as an analysis of its economic feasibility. According to FVM, the more viable a technology for the firm, the more readiness firm to use the technology, the better the firm's performance. Thus, it can be proposed:

H8. The viability of sCRM in the context of SMEs positively impacts SMEs' performance.

3.9. Performance

The company's performance reflects its effectiveness in management and the value it delivers to customers and other stakeholders. It signifies a manager's perception of how the company outperforms its competitors on multiple metrics (Demirbag, Koh, Tatoglu & Zaim, 2006; Tarutė & Gatautis, 2014). In the context of this study, performance refers to the real adventures gained by SMEs from utilizing social CRM in terms of both financial (sales transactions and sales volume) and non-financial returns (sales inquiries, number of customers, and brand exposure) (Ainin, Parveen, Moghavvemi, Jaafar & Shuib, 2015; Odoom, Anning-Dorson & Acheampong, 2017). As discussed earlier, according to FVM, performance can be enhanced when the firm adopts a technology that fits the firm's requirements (fitness), and the firm is capable of implementing the technology (viability).

4. Methodology

In this study, a positivist research paradigm was employed for its focus on quantitative data, facilitating the efficient measurement of responses on a large scale across SMEs in Malaysia. The questionnaire was deemed a suitable tool for conducting comprehensive investigations into variable relationships and testing hypotheses. Furthermore, the survey enables researchers to assess the attitudes and decisions of respondents related to the phenomenon under consideration. Accordingly, an online survey was designed to obtain quantitative data from SMEs' management personnel to evaluate our study model. Questionnaires were distributed using a simple random sampling technique. Due to the lockdown during the COVID-19 pandemic, non-probability sampling through email was utilized. Among more than 3000 emails and messages with the questionnaire distributed across various platforms, we obtained 216 replies resulting in a response rate of 7%. Further visual filtration leads to only 149 complete and usable responses for data analysis. The data collection was carried out between September 2020 and October 2021. The weak response rate was predicted during the lockdown at the time of COVID-19 pandemic. In addition, the rising volume of online surveys led to survey weariness. Furthermore, the pandemic stresses SME owners/managers, causing them to experience high psychological pressure, making it harder for this study to reach the optimal respondent number (Baghdadi, Younis, Al Suwaidan, Hassounah & Al Khalifah, 2021). The instrument used to collect the data is illustrated in the following sections.

4.1. Instrument development

In this study, a specialized tool or instrument has been created to collect data and examine the proposed model, aiming to identify the essential factors that impact the fit and viability of sCRM on SMEs performance. To create this instrument, the study carefully selects the variables that constitute the proposed model and then operationalizes

Table 1
The construct operational definitions.

The construct	Operational Definition
Internal Financial Resources	Financial investment is required to customize the chosen social media solution to suit the company's specific requirements and to provide training for employees to effectively operate the sCRM system.
Governmental Support	The policies, programs, incentives, and agencies initiated by the government encourage the adoption of new technology or innovations.
Manager Innovativeness	The readiness of the managers to accept new technology and deliver new experiences to their firms.
IT/IS Knowledge of Employees	Possessing sufficient expertise in utilizing social media for Customer Relationship Management (CRM)
Top Management Support	The extent to which senior management may create a fruitful atmosphere for sCRM adoption by providing resources.
Viability	The degree to which SMEs are prepared to adopt social media and the potential value it can add to their Customer Relationship Management (CRM) efforts.
Network externalities	To what degree is Customer Relationship Management (CRM) correlated with the number of customers who utilize social media tools?
Fit	To what extent do the characteristics of social media align with the specific needs of a task or tasks intended to achieve CRM objectives.
Performance	– A measure for evaluating the effectiveness of SMEs' management and the value they deliver to customers and other stakeholders. – Managers' opinions of how the company fared on a variety of organizational performance metrics in comparison to its rivals; and – The real advantages SMEs gain from adopting social media in terms of financial and non-financial performance.

them according to the unique context of the research. These operationalizations are outlined in detail in Table 1, providing an overview of the variables and their measurements used in the study. Then, items are developed and derived from original measurements built based on their original theory, applied in various research areas in the literature, and proved valid and rigorous. These research practices help establish the instrument's validity (Agarwal, 2011). Further, to enhance the instrument's validity, content validity is conducted by experts in the field.

To ensure the content validity of the instrument used in the current study, three experienced researchers were invited to evaluate whether the items representing the various dimensions of the constructs adequately covered the full range of possible measures. During this examination, it was discovered that two items lacked sufficient significance in their wordings. As a result, minor revisions were made based on the suggestions provided by the evaluators. Additionally, one item was eliminated from further consideration as it was deemed not sufficiently relevant. Furthermore, to enhance the instrument's comprehensiveness, two items were added to the constructs of "Internal Financial Resources" and "Government Support." This addition was made to ensure that each construct contained at least three items, a recommendation provided by the experts. These new items were carefully derived and developed from relevant literature, considering the specific context of SMEs' environment.

By undergoing this rigorous validation process, the instrument's content validity was strengthened, ensuring its suitability and accuracy for capturing the data necessary for the study's investigation.

5. Analysis and results

5.1. Descriptive statistic

Table. 2 demonstrates the mean and Standard Deviation for all measurements in the instrument. All the indicators were measured using 5 points Likert scale from 1 to 5 (from Strongly Disagree to Strongly Agree). As can be seen in Table. 2, the mean of performance scale ranges from 3.857 (for P1) with a Standard Deviation (SD) of 0.683 to 4.476 (for P5) with an SD of 0.707. This indicates that the respondents assign high scores for the impact of sCRM on the performance of SMEs. In addition, for the fit scale, all have a mean over 3 (3.881 to 4.405) with an SD range from 0.670 to 0.916, while the scale for viability has a mean of 3.476 and 4.00 and an SD of 0.692 and 0.733. This indicates that respondents support the fitness of sCRM with the SME's requirements, and it is viable to be implemented to enhance the SMEs' performance.

5.2. Measurement model (Outer model)

In the Partial Least Squares (PLS) analysis, the initial step involves analyzing the Measurement Model to assess how well the indicators (items) have successfully loaded onto their respective theoretically defined constructs. By examining the outer model, researchers can ensure that the survey indicators accurately measure the constructs they were intended to represent. This process is critical for establishing the reliability and validity of the survey instrument, thus enhancing the overall quality of the research outcomes.

I. *Convergent Validity*

Convergent validity indicates that a measure correlates positively with other measures in the same construct, while discriminant validity is an indicator of the distinction of a construct from other constructs. Outer loadings and the Average Variance Extracted (AVE) are indicators of convergent validity. Results showed that the outer loading for IT2 with 0.233 and TMS1 with 0.251 are less than the rule of thumb (0.5). Thus, they were removed. Table. 3 and Fig. 1 illustrate the factor loading for all constructs. Furthermore, all AVEs are more than 0.5, which is a rule of thumb for a satisfactory level of convergent validity (Alzahrani, Stahl & Prior, 2012; Hair Jr, Hult, Ringle & Sarstedt, 2014) (see Table. 3) .

II. *Discriminant Validity*

Discriminant validity refers to the extent to which measures of a specific construct do not correlate significantly with measures of other constructs that are not intended to assess the same underlying construct (Hair, Hult, Ringle & Sarstedt, 2014). To assess the discriminant validity of all the constructs in the model, the researchers utilize the Fornell-Larcker criterion. Fornell and Larcker (Fornell & Larcker, 1981), the Average Variance Extracted (AVE) values for each construct should be greater than the squared correlations between that construct and all other constructs in the model.

In Table. 4, The researchers reveal the outcomes of their analysis, demonstrating that the square root of the Average Variance Extracted ($\sqrt{AVE}$) for each construct is larger than the correlations between that specific construct and all other constructs in the model. This finding indicates that the constructs possess discriminant validity, as they are distinct from one another and do not overlap significantly. The study successfully establishes the discriminant validity of the constructs (Alzahrani et al., 2012), ensuring that each construct effectively measures a unique aspect of the research model without being unduly influenced by other constructs.

Discriminant validity can also be assessed by using the Heterotrait-

Table 2
Means and Standard Deviations (SD).

Construct	Item	Mean (M)	SD	Construct	Item	Mean (M)	SD
FIT	FIT1	4.405	0.701	NETWORK EXTERNALITIES	NE1	3.929	0.778
	FIT2	3.881	0.670		NE2	3.976	0.715
	FIT3	4.071	0.712		NE3	3.905	0.726
	FIT4	4.048	0.764		NE4	4.238	0.656
	FIT5	4.024	0.841	SMES PERFORMANCE	P1	3.857	0.683
	FIT6	3.881	0.916		P2	3.881	0.670
GOVERNMENTAL SUPPORT	GS1	3.667	0.928		P3	4.310	0.749
	GS2	3.405	1.014		P4	4.214	0.717
	GS3	3.405	0.885		P5	4.476	0.707
IT/IS KNOWLEDGE OF EMPLOYEES	IT1	3.762	0.790	TOP MANAGEMENT SUPPORT	TMS1	4.333	0.612
	IT3	3.595	0.828		TMS2	4.310	0.604
INTERNAL FINANCE RESOURCES	IFR1	3.690	0.924		TMS3	4.286	0.742
	IFR2	3.571	0.887		TMS4	4.286	0.708
	IFR3	3.500	0.890		TMS5	4.048	0.882
MANAGER INNOVATIVENESS	MI1	4.262	0.828	VIABILITY	V1	3.762	0.692
	MI2	4.333	0.687		V2	3.476	0.707
	MI3	3.929	0.838		V3	4.000	0.733

Table 3
Factor Loadings and AVE for Model Constructs.

Construct	AVE	Indicator	Loading	Construct	AVE	Indicator	Loading
FIT	0.569	FIT1	0.645	NETWORK EXTERNALITIES	0.752	NE1	0.816
		FIT2	0.621			NE2	0.906
		FIT3	0.673			NE3	0.900
		FIT4	0.844			NE4	0.842
		FIT5	0.791	SMES PERFORMANCE	0.698	P1	0.894
		FIT6	0.905			P2	0.909
GOVERNMENTAL SUPPORT	0.735	GS1	0.788			P3	0.825
		GS2	0.941			P4	0.868
		GS3	0.836			P5	0.655
IT/IS KNOWLEDGE OF EMPLOYEES	0.656	IT1	0.997	TOP MANAGEMENT SUPPORT	0.528	TMS2	0.628
		IT3	0.565			TMS3	0.753
INTERNAL FINANCE RESOURCES	0.907	IFR1	0.968			TMS4	0.856
		IFR2	0.957			TMS5	0.941
		IFR3	0.932	VIABILITY	0.600	V1	0.832
MANAGER INNOVATIVENESS	0.559	MI1	0.464			V2	0.602
		MI2	0.877			V3	0.863
		MI3	0.833				

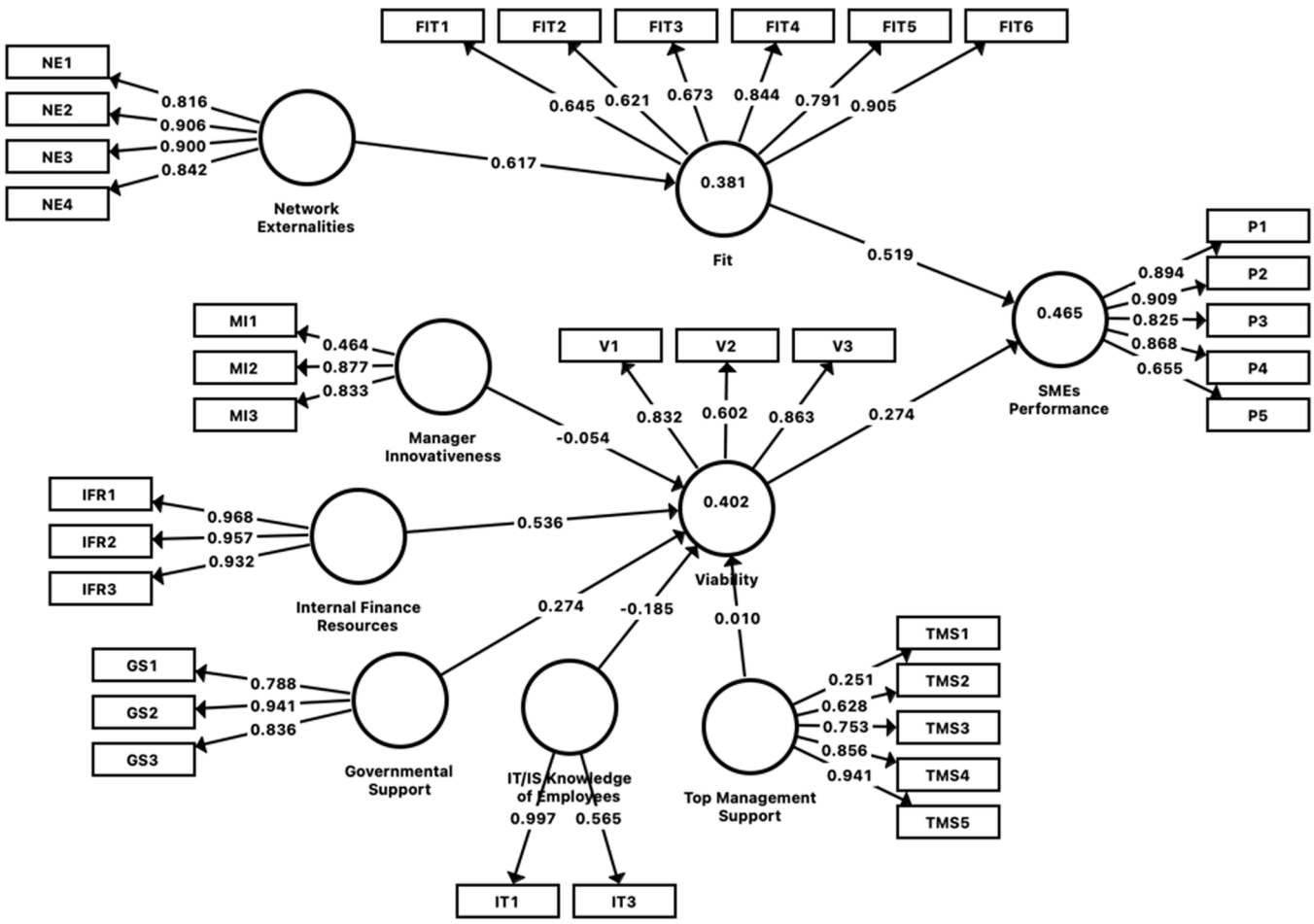

Fig. 1. Factor loadings for all indicators.

Monotrait Ratio (HTMT) (Henseler, Ringle & Sarstedt, 2015). According to Henseler, et al. (Henseler et al., 2015), if the values are close to 1.0, the constructs are not perfectly correlated, which indicates a lack of discriminant validity. The results presented in Table. 5 show that the discriminant validity of all the constructs has been established (Gaskin, Godfrey & Vance, 2018; Henseler et al., 2015).

III. *Reliability*

To assess the reliability of the measures, two indicators, Composite Reliability (CR) and Cronbach's Alpha, are used. According to the recommendation Hair Jr, et al. (Hair Jr et al., 2014) a Composite Reliability value of at least 0.70 is considered reliable for all variables. The results presented in Table. 6 indicate that all variables meet this recommended value of 0.70, confirming their reliability. Similarly, the study examines Cronbach's Alpha, which should exceed a minimum threshold of 0.6 (Hair et al., 2014) or all constructs. In Table. 6, it is evident that Cronbach's Alpha values for all constructs surpass this

Table 4
Fornell-Larcker Criterion.

	Fit	Governmental Support	IT/IS Knowledge of Employees	Internal Finance Resources	Manager Innovativeness	Network Externalities	SMEs Performance	Top Management Support	Viability
Fit	**0.754**								
Governmental Support	0.099	**0.857**							
IT/IS Knowledge of Employees	0.179	−0.034	**0.606**						
Internal Finance Resources	0.360	−0.094	−0.080	**0.953**					
Manager Innovativeness	−0.197	−0.020	−0.077	−0.252	**0.745**				
Network Externalities	0.617	0.326	0.164	0.447	−0.221	**0.867**			
SMEs Performance	0.635	0.116	0.127	0.431	−0.169	0.510	**0.835**		
Top Management Support	0.031	0.251	0.060	0.224	0.040	0.369	0.218	**0.733**	
Viability	0.416	0.222	−0.240	0.539	−0.189	0.299	0.489	0.171	**0.776**

Table 5
Heterotrait-Monotrait Ratio (HTMT).

	Fit	Governmental Support	IT/IS Knowledge of Employees	Internal Finance Resources	Manager Innovativeness	Network Externalities	SMEs Performance	Top Management Support	Viability
Fit									
Governmental Support	0.274								
IT/IS Knowledge of Employees	0.384	0.361							
Internal Finance Resources	0.435	0.101	0.118						
Manager Innovativeness	0.344	0.160	0.467	0.336					
Network Externalities	0.685	0.414	0.350	0.489	0.349				
SMEs Performance	0.662	0.231	0.296	0.437	0.285	0.559			
Top Management Support	0.226	0.330	0.469	0.146	0.587	0.390	0.278		
Viability	0.600	0.325	0.419	0.682	0.357	0.418	0.624	0.140	

Table 6
Composite Reliabilities of Constructs in Model.

Construct	Composite Reliability	Cronbach's Alpha
Fit	0.886	0.845
Governmental Support	0.892	0.822
IT/IS Knowledge of Employees	0.780	0.774
Internal Finance Resources	0.967	0.949
Manager Innovativeness	0.781	0.661
Network Externalities	0.924	0.890
SMEs Performance	0.919	0.894
Top Management Support	0.833	0.889
Viability	0.815	0.802

required threshold, further affirming the reliability of the measurement model.

Based on these findings, the measurement model provides sufficient evidence of reliability, ensuring that the measures used in the study are dependable and consistent in capturing the intended aspects of the constructs (Alzahrani et al., 2012).

5.3. Structural model results (Inner model)

After analyzing the Measurement Model in PLS, the structural model is generated by examining the inner model to test the significance of the hypotheses proposed in the study. The first step in this process involves identifying the significance of the paths between constructs, which is done using computed T-statistics. For a hypothesis to be considered significant, the T-statistics value should be at least 1.96.

To test the significance of the model's hypotheses, the researchers conducted 5000 bootstrapped samples with 0 cases per sample. The results of the T-statistics and corresponding p-values are presented in Table. 7. These results enable the identification of the significance of the model's hypotheses and provide evidence regarding the relationships between different constructs in the proposed model.

By employing this rigorous analysis, the study ensures that the structural model's hypotheses are supported by statistically significant findings, validating the proposed relationships between the constructs, and contributing to the overall robustness of the research outcomes.

Results from the Bootstrapping analysis, as shown in Fig. 2, the hypothesis' validation is explained as the following:

Hypothesis 1. The value of t-statistics for Network Externalities towards Fit is 4.533, and the p-value is 0.000. Therefore, Network Externalities have a significant positive effect on Fit.

Hypothesis 2. The value of t-statistics for Manager Innovativeness towards Viability is 0.301, and the p-value is 0.763. Therefore, the effect of Manager Innovativeness on Viability is not significant.

Hypothesis 3. The value of t-statistics for IT/IS Knowledge of Employees towards Viability is 0.927, and the p-value is 0.354. Therefore, IT/IS Knowledge of Employees doesn't significantly affect Viability.

Hypothesis 4. The value of t-statistics for Top Management Support

Table 7
T-statistics and p-values.

Hypotheses	Original Sample	Sample Mean	Standard Deviation	T Statistics	P Values
Network Externalities -> Fit	0.617	0.612	0.136	4.533	0.000
Manager Innovativeness -> Viability	−0.054	−0.089	0.180	0.301	0.763
IT/IS Knowledge -> Viability	−0.185	−0.101	0.199	0.927	0.354
Top Management Support -> Viability	0.010	0.027	0.158	0.066	0.948
Internal Finance Resources -> Viability	0.536	0.499	0.146	3.680	0.000
Governmental Support -> Viability	0.274	0.252	0.176	1.556	0.120
Fit -> SMEs Performance	0.519	0.527	0.188	2.756	0.006
Viability -> SMEs Performance	0.274	0.268	0.172	1.596	0.110

*$p < 0.05$.

towards Viability is 0.066, and the p-value is 0.948. Therefore, Top Management Support doesn't affect the viability of sCRM for SMEs.

Hypothesis 5. The value of t-statistics for Internal Finance Resources towards Viability is 3.680, and the p-value is 0.000. Thus, Internal Finance Resources significantly affect the viability of sCRM for SMEs.

Hypothesis 6. The value of t-statistics for Governmental Support towards Viability is 1.556, and the p-value is 0.120. Thus, Governmental Support has a non-significant effect on the viability.

Hypothesis 7. The value of t-statistics for Fit towards SMEs Performance is 2.756, and the p-value is 0.006. Therefore, Fit has a significant impact on SMEs' Performance.

Hypothesis 8. The value of t-statistics for Viability toward SMEs Performance is 1.596, and the p-value is 0.110. Therefore, Viability doesn't have a significant impact on SMEs' Performance.

In addition, Results show that the coefficient of determination, R^2, is 0.465 for the SMEs Performance construct. This means that the construct explains 46.5% of SME performance variance.

6. Discussion

The primary objective of this study was to offer a well-founded conceptual framework to understand the implications of incorporating social media into CRM strategies and implementations on SMEs performance. Social media's network externality feature has the potential to make it an ideal platform for CRM tasks. However, to effectively integrate social media tools with the CRM system and enhance the company's performance, a thorough analysis of the organization's internal resources is necessary. This analysis helps identify the firm's capabilities and aids in devising a strategy to achieve sustainable

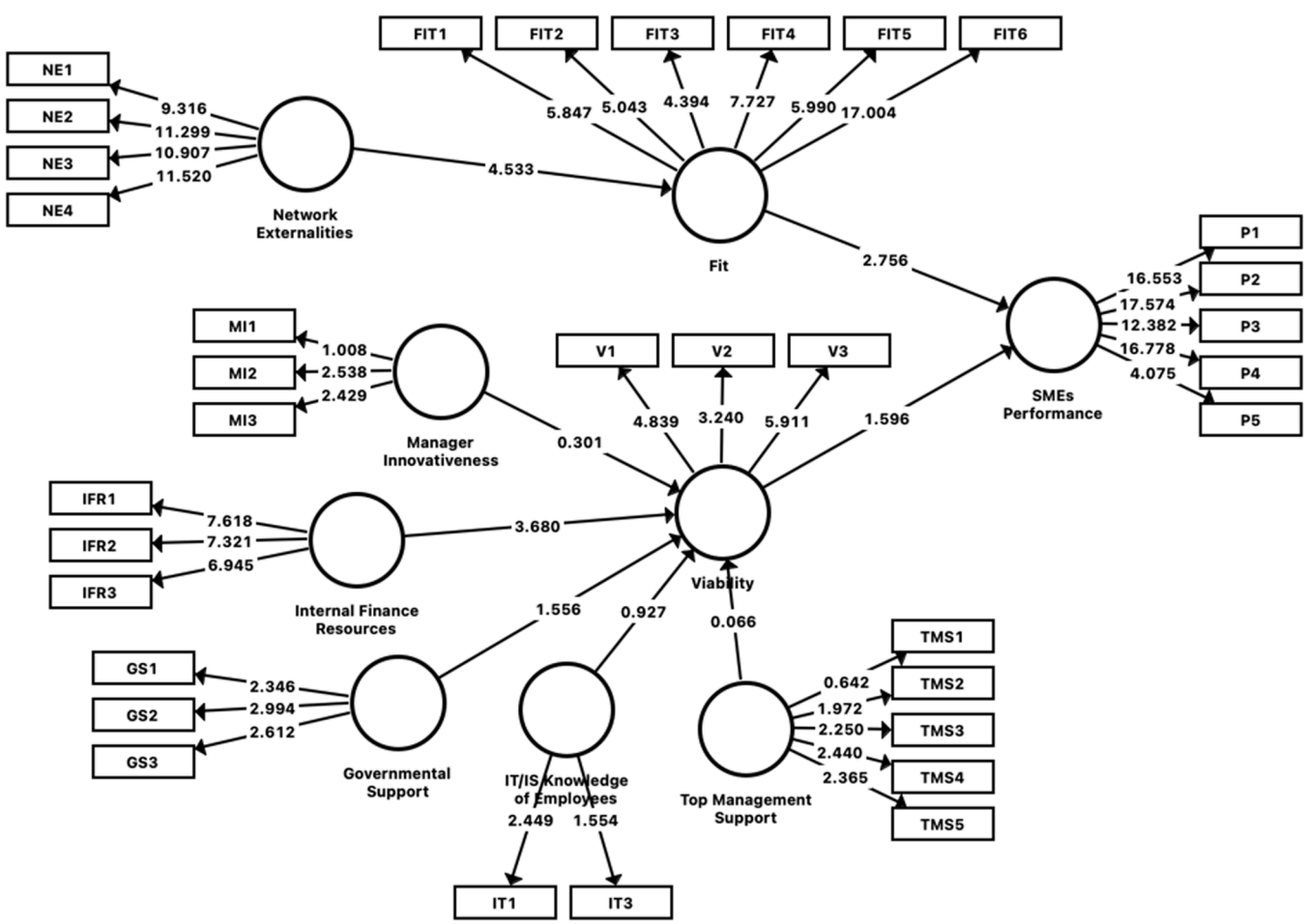

Fig. 2. T Statistics.

competitive advantages through the successful integration of social media into their CRM approach. By providing a theoretical foundation for this integration, the study contributes valuable insights into optimizing CRM strategies and implementations for SMEs, ultimately leading to improved performance and competitive positioning in the market.

The results of our study reveal significant insights into the factors influencing the suitability of social CRM for SMEs. Firstly, our findings underscore the pivotal role of network externalities in shaping the fitness of sCRM to SMEs' requirements. This is evidenced by the statistically significant relationship supported by a high t-value ($t = 4.533$) and a significant p-value ($P = 0.000$). The results suggest that the increased engagement of customers on social media platforms contributes to the enhanced value proposition of sCRM, aligning with the notion that social media facilitates broad customer engagement (A.O. Banwo, Du & Onokala, 2017; V. Gu, Davis, Cao & Vogt, 2017). Moreover, our analysis indicates a noteworthy association between internal financial resources and the viability of sCRM adoption among SMEs (Hypothesis 5). The empirical evidence confirms this relationship, with a notable t-value of 3.680 and a p-value of 0.000. This suggests that despite resource constraints inherent to SMEs, leveraging social media as a cost-effective CRM tool remains feasible, aligning with prior research (Kirtiş & Karahan, 2011; Kurnia et al., 2015; Tan et al., 2009).

However, our study also reveals non-significant relationships between manager innovativeness ($t = 0.301$, $P = 0.763$), IT knowledge ($t = 0.927$, $P = 0.354$), and top management support ($t = 0.066$, $P = 0.948$) with the viability of sCRM for SMEs. These findings diverge from previous literature (Ahani et al., 2017; Ahmad et al., 2019; Beier & Wagner, 2016; Effendi et al., 2020; Hasani et al., 2017; Park et al., 2015; Sugandini et al., 2020), which predominantly focused on social media adoption for marketing rather than CRM purposes in SME contexts and the impact on the performance.

Additionally, while the relationship between government support (Hypothesis 6) and sCRM viability did not achieve statistical significance ($t = 1.556$, $p = 0.120$), the proximity of the t-value to the threshold of significance warrants further exploration, particularly considering its potential implications for policy interventions supporting SME digitalization efforts. Furthermore, our investigation into the impact of viability and fit on performance revealed significant insights. While the hypothesis pertaining to viability-performance relationship (Hypothesis 8) narrowly missed significance ($t = 1.596$, $p = 0.110$), suggesting a potential trend warranting further inquiry, the significant association between fit and performance (Hypothesis 7) ($t = 2.756$, $p = 0.006$) underscores the pivotal role of aligning sCRM with SMEs' specific needs and operational contexts. These findings in line with the FVM and related literature which examined the fitness of different technologies in different contexts (Larosiliere & Carter, 2016; Liang et al., 2021; Tripathi & Nasina, 2017), emphasizing the importance of comprehensively assessing technological fit and economic feasibility when implementing sCRM within SME environments.

The findings of this study align with the existing literature concerning the influence of adopting technological innovations on performance. It highlights the idea that merely adopting technology in isolation may not be adequate to attain a competitive advantage. The literature suggests that the successful integration and utilization of technology, such as social media in CRM strategies, requires a more comprehensive approach. It involves considering various factors, including internal resources, organizational capabilities, and the formulation of effective strategies. By acknowledging the importance of these aspects, the study reinforces the understanding that technology is just one piece of the puzzle and that leveraging it effectively within the broader organizational context is crucial for gaining a competitive edge and achieving improved performance outcomes (Baas & Rekom, 2010; Goodhue & Thompson, 1995; Ikechukwu, 2019; T.P. Liang et al., 2007; Tjan, 2001).

7. Limitations and future works

Despite the valuable insights provided by our integrated model, several limitations warrant consideration. Firstly, the reliance on cross-sectional survey data limits our ability to establish causal relationships between variables. Future research could employ longitudinal or experimental designs to provide stronger evidence of causality. Secondly, although efforts were made to obtain a representative sample of SMEs managerial staff, the sample size of 149 participants may limit the generalizability of our findings to broader populations of SMEs. Future studies could aim for larger and more diverse samples to enhance the external validity of the research. Thirdly, this study focuses primarily on SMEs in developing countries, which may restrict the applicability of our findings to SMEs operating in different contexts or regions with varying levels of technological infrastructure and market dynamics. Future research could explore the nuances of social CRM adoption in diverse settings to provide a more comprehensive understanding of its impact. Finally, while our integrated model incorporates prominent theoretical frameworks, such as the Fit-Viability Model, Network Externalities, and the Resource-Based View, there may be other relevant theories or factors not accounted for in our analysis that could influence the relationship between social CRM adoption and SME performance.

Recognizing these limitations, future research should employ longitudinal designs and larger, more diverse samples to further elucidate the causal relationships between social CRM adoption and SME performance. Additionally, exploring additional contextual factors and employing mixed-methods approaches could provide a more comprehensive understanding of the dynamics at play.

8. Conclusion

In this study, an integrated model is presented, which identifies the critical factors to be considered while exploring the impact of social Customer Relationship Management (CRM) on the performance of Small and Medium-sized Enterprises (SMEs), especially in developing countries. This model amalgamates two theories, Network Externalities and Resource-Based View (RBV), with the Fit Viability Model (FVM). The firm's performance is influenced by two primary factors: the alignment of social media characteristics with CRM tasks (fitness) and the firm's preparedness to implement sCRM (viability). By employing the proposed model, SMEs in developing nations can make well-informed decisions regarding the adoption of social media tools to effectively manage their customer relationships based on their specific requirements and capabilities.

Data was collected to empirically test the proposed model. Consequently, this study makes a valuable contribution to the current literature on the influence of technology on SMEs' performance by providing a comprehensive perspective that incorporates two crucial aspects: organizational readiness, guided by the RBV theory, and technology characteristics, based on the Network Externalities theory. These theories are seamlessly integrated with FVM, serving as the foundational theory of the proposed sCRM impact model on SMEs' performance. By providing this holistic insight, the study emphasizes the positive impact of integrating social media tools for CRM on firm performance, taking into account the firm-level capabilities and internal resources of SMEs.

In conclusion, the proposed model not only provides a framework for future conceptual and empirical research on social CRM but also substantiates the argument that integrating social media tools into CRM practices can genuinely improve firm performance. By emphasizing the importance of considering internal capabilities and technology characteristics, the study provides valuable guidance for SMEs seeking to leverage sCRM effectively to achieve a competitive advantage and improved overall performance.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve the clarity and the readability of our writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Ethical approval

All the authors mentioned in the manuscript have agreed for authorship, read and approved the manuscript, and given consent for submission and subsequent publication of the manuscript.

CRediT authorship contribution statement

Fathey Mohammed: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Rahayu Binti Ahmad:** Conceptualization, Formal analysis, Methodology, Validation. **Syahida Binti Hassan:** Data curation, Methodology, Project administration, Resources, Software. **Yousef Fazea:** Formal analysis, Investigation, Resources, Writing – review & editing. **Ahmed Ibrahim Alzahrani:** Funding acquisition, Investigation, Resources, Software, Writing – review & editing.

Declaration of competing interest

The authors declare they have no conflicts of interest to report regarding the present study.

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